Physical Cume
April 3, 2009

(1) (50 points) Prove that, for a two-component solution at given T: $\mathbf{x_1 d(\ln\gamma_1) + x_2 d(\ln\gamma_2) = 0}$. Here, x_1 and x_2 are the molar fraction for the two components, and γ_1 and γ_2 are the activity coefficients for the two components.

(2) (50 points) Prove: $\mathbf{\Delta_r H^0 = -\nu F[E^0 - T(dE^0/dT)]}$. $\Delta_r H^0$ is the stadnard reaction enthalpy for a given redox reaction. F is the Faraday constant. $\mathbf{E^0}$ is the corresponding standard electrochemical potential of the redox reaction. T is the absolute temperautre. $\boldsymbol{\nu}$ is the number of electrons trnasferred between the reductant and oxidant of the reaction.